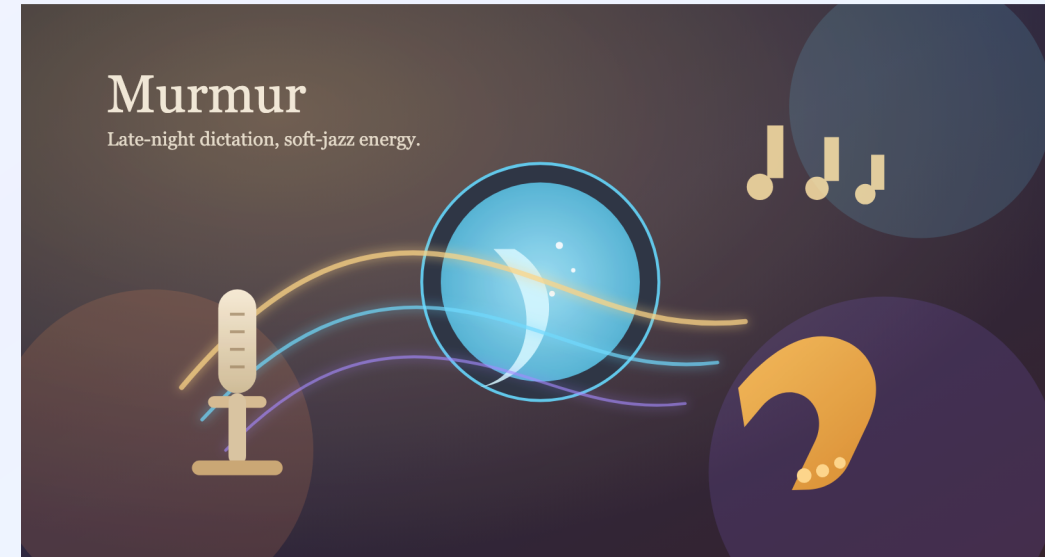


# Murmur

## Local-first dictation that feels instant

### What this talk covers

- System architecture
- Moonshine Voice: what it is and how it works
- Cleaning pipeline and optional rewrite model



# Agenda

1. Product experience in one minute
2. Architecture and reliability design
3. Moonshine Voice deep dive
4. Cleaning pipeline and rewrite safety
5. Demo flow

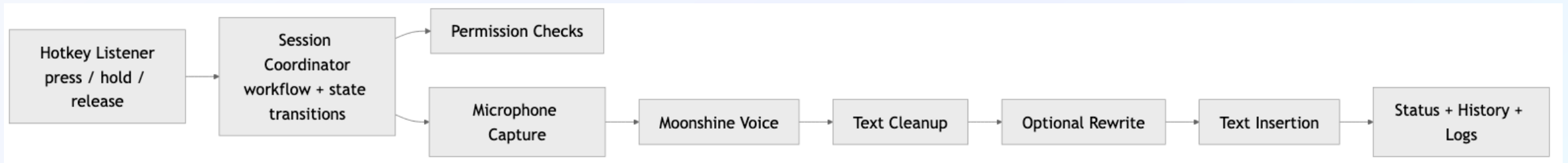
# The Product Loop

Hold shortcut -> Speak -> Release -> Text appears

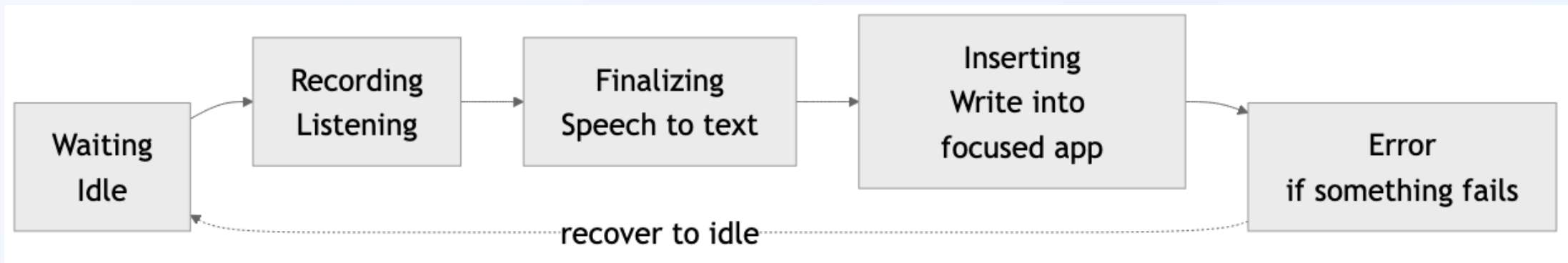
Why this is hard:

- Must feel fast
- Must work across apps
- Must recover safely when something fails

# Murmur Architecture (Visual)



# Runtime Reliability Model



## Moonshine Voice: What It Is

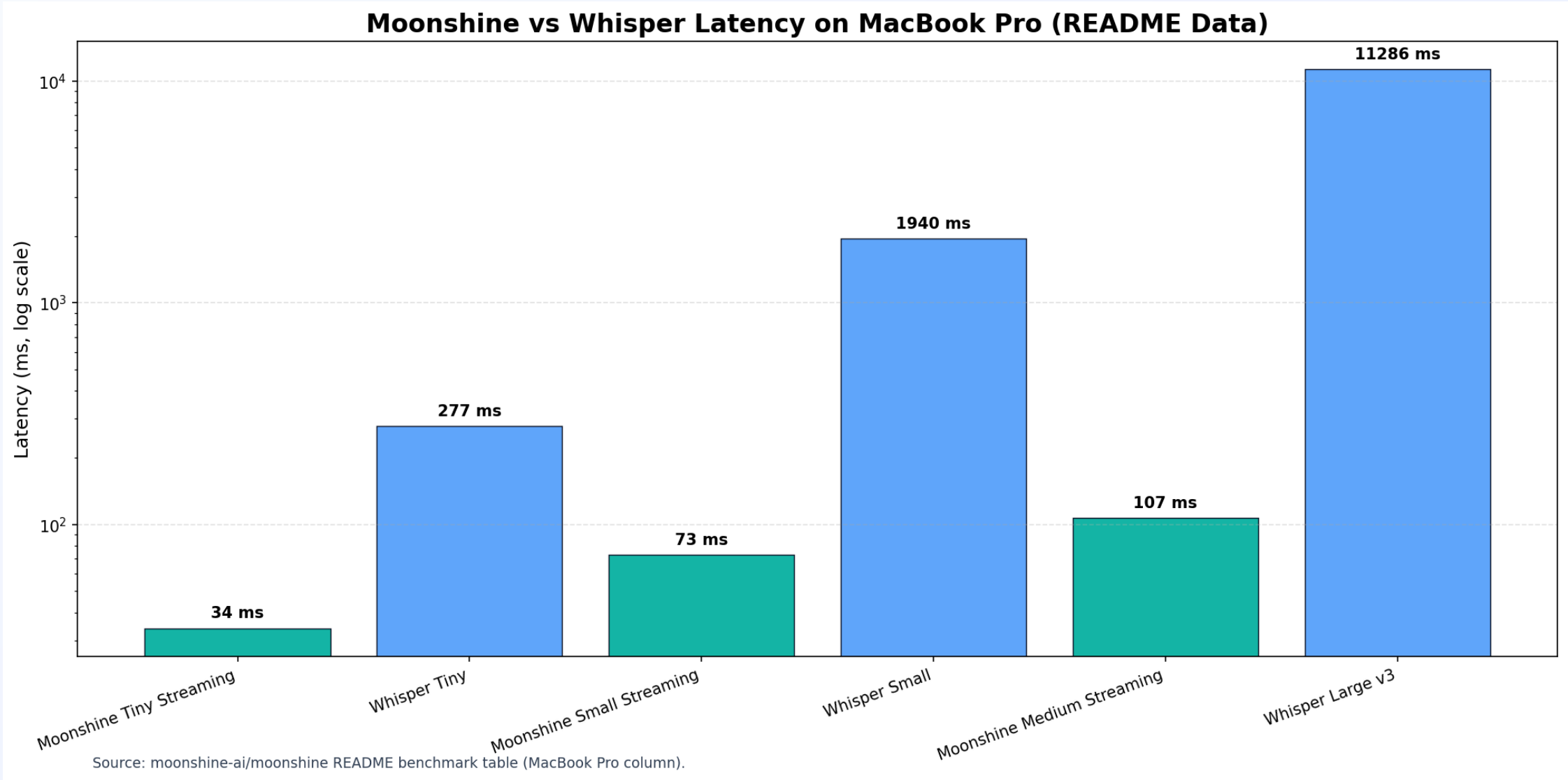
- Open-source toolkit for live voice interfaces
- Designed for on-device speech recognition
- Optimized for low-latency streaming use cases

# Moonshine Voice: How It Works

- Audio enters through microphone capture
- Voice activity detection segments speech in real time
- Speech-to-text + intent stages produce app actions

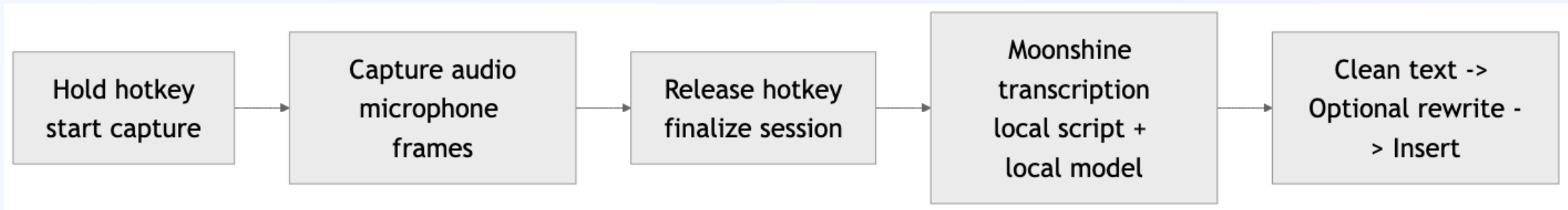
Source: Moonshine open-source architecture diagram.

# Why We Chose Moonshine for Murmur

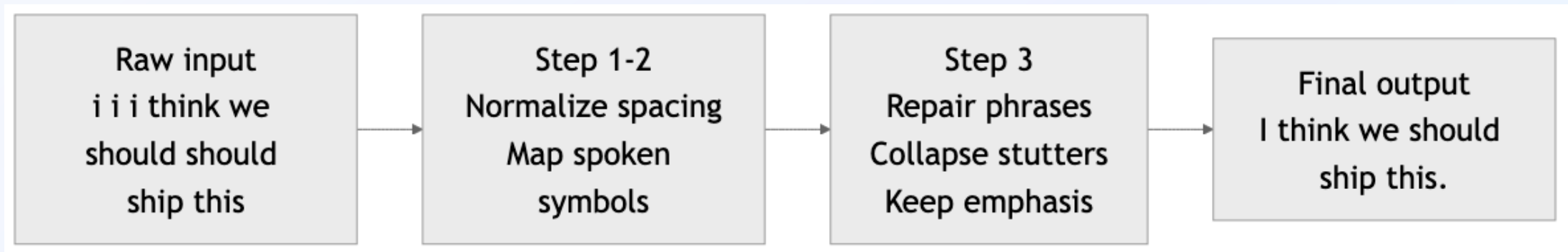




## How Murmur Uses Moonshine in Practice



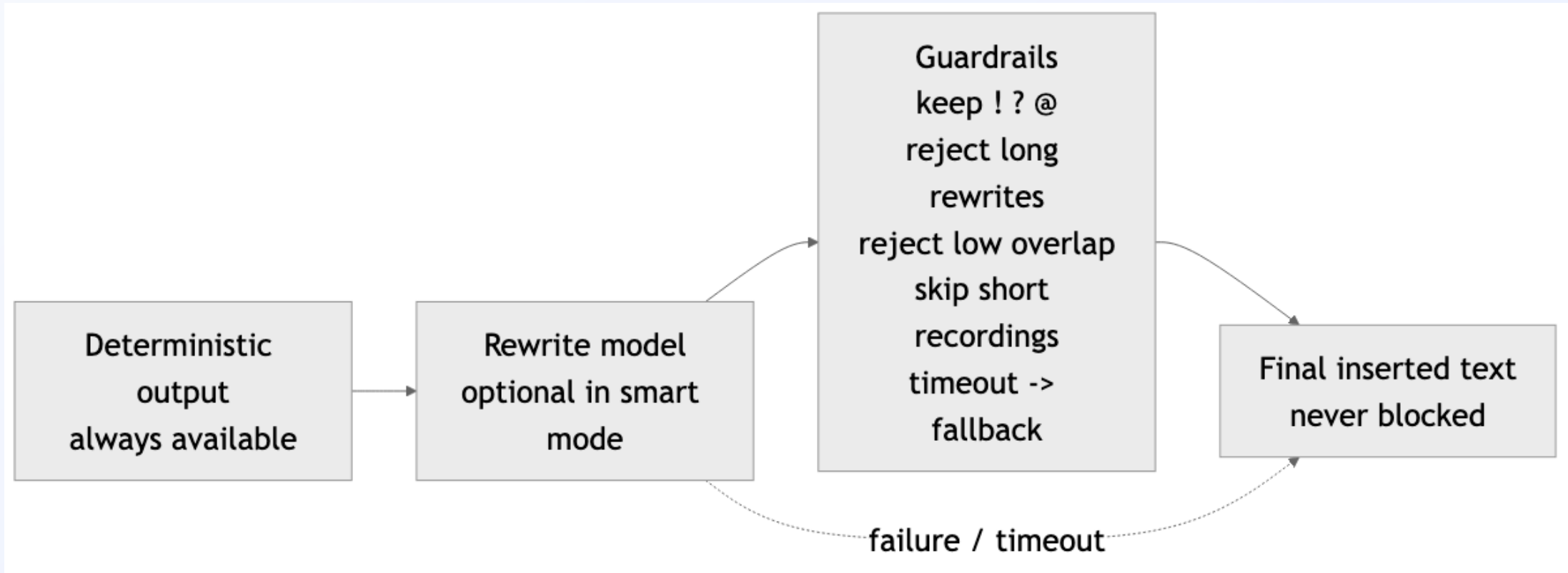
# Deterministic Cleaning Pipeline



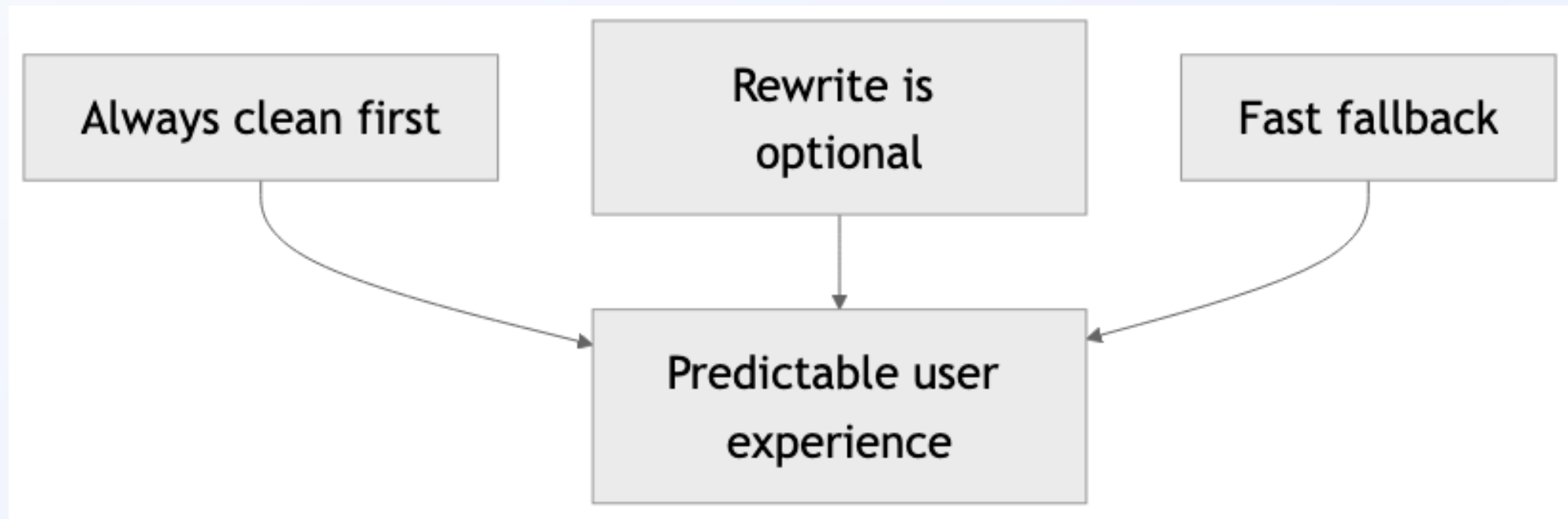
## Cleaning Examples (Before -> After)

- `i i i think we should should ship this`  
-> `I think we should ship this.`
- `send it tomorrow scratch that send it friday`  
-> `Send it friday.`
- `send this to jane at sign example.com`  
-> `Send this to jane@example.com.`

# Optional Rewrite Model With Guardrails



## What "Safe by Default" Means Here



## Demo Plan (Live)

1. Show press-hold-release insertion loop
2. Show one noisy input cleaned to stable text
3. Toggle deterministic mode vs smart mode
4. Show logs for timing and insertion path

## Takeaways

- Murmur is designed as a reliable real-time system
- Moonshine gives a strong local speech-recognition backbone
- Deterministic cleanup provides predictable quality
- The rewrite layer adds polish without risking core behavior

**Q&A**

Thanks.